

Lead Quality Analysis

DebtReductionInc Case Study — Lead Quality, Trends & Drivers

Dataset: 3,021 leads sold to one advertiser, April–September 2009, sourced from one publisher across five traffic partners and 12 ad creative variants.

Executive Summary

Across the 3,021 leads in this dataset, **8.1% closed** — matching the advertiser's reported 8.0% baseline. The three questions below are answered in turn:

- **Q1 — Trend:** Yes, there is a statistically significant trend, but it is not a simple decline or improvement. Quality fell sharply from April through July, then recovered through September (a “U-shape,” $p < 0.0001$). Much of this swing tracks a shift in traffic-source mix, not a structural decline in lead quality.
- **Q2 — Drivers:** The clearest, most actionable driver is self-reported **debt level** — quality peaks for mid-range debt (\$10k–\$90k) and drops off sharply at both extremes ($p = 0.006$). **Traffic source** is the second lever: AdKnowledge and the Call Center channel meaningfully outperform Google and Yahoo. Address/phone verification scores show a promising but not yet conclusive signal. Ad creative, form length, and online-vs-call-center entry showed *no* statistically detectable effect on quality.
- **Q3 — Reaching 9.6%:** Yes — there is real headroom. Simply excluding leads outside the \$10k–\$90k debt sweet spot would have lifted the historical closed rate from 8.1% to **9.0%** on its own. Layering in traffic-mix optimization toward AdKnowledge/Call Center pushes a representative subset of this data to **11.3%** — comfortably past the 9.6% target. Recommendations are detailed in Section 3.

A Note on Method & Data Caveats

Lead quality metric. The advertiser's bottom-line measure is the *Closed* rate (lead became a paying customer) — this is what the 8.0%/9.6% figures in the prompt reference, and what this report uses as the primary KPI. Where a larger, more statistically powered view is useful, this report also reports a broader “Good lead” rate that adds in leads still progressing through the EP Sent / EP Received / EP Confirmed pipeline — stages the advertiser itself treats as positive quality signals, short of a close.

Right-censoring. As of the data pull, 71% of leads (2,140 of 3,021) had no CallStatus at all — i.e., still pending (“Unknown” in this report). A lead can sit in this state indefinitely (e.g., a worksheet never returned), and leads created later in the window have had less calendar time to resolve. This means raw month-over-month Closed rates can look worse for recent months simply because fewer of those leads have had time to mature — not because they are lower quality. Throughout Section 1, trend figures are shown both ways (raw, and restricted to leads that have already resolved to a final status) so this effect is visible rather than hidden.

AddressScore / PhoneScore. These fields were only captured starting in July (0% populated April–June, 74–99% populated July–September). Findings using these fields are based on that smaller, more recent subset and should be treated as directional pending a longer track record.

Statistical testing. Chi-square tests of independence and logistic regression are used throughout; results are reported with p-values. A result is called “significant” at the conventional $p < 0.05$ threshold, with effect sizes (percentage-point gaps, sample sizes) always shown alongside so the practical magnitude is clear, not just statistical detectability.

1. Are We Seeing a Lead-Quality Trend Over Time?

Short answer: **yes, a significant trend exists, but it's a dip-and-recovery, not a simple drift in one direction.**

1.1 What the raw data shows

Looking at every lead by the month it was created, the share that's still unresolved ("Unknown") dominates every month — a reminder that this dataset is a snapshot of an in-flight pipeline, not a closed book.

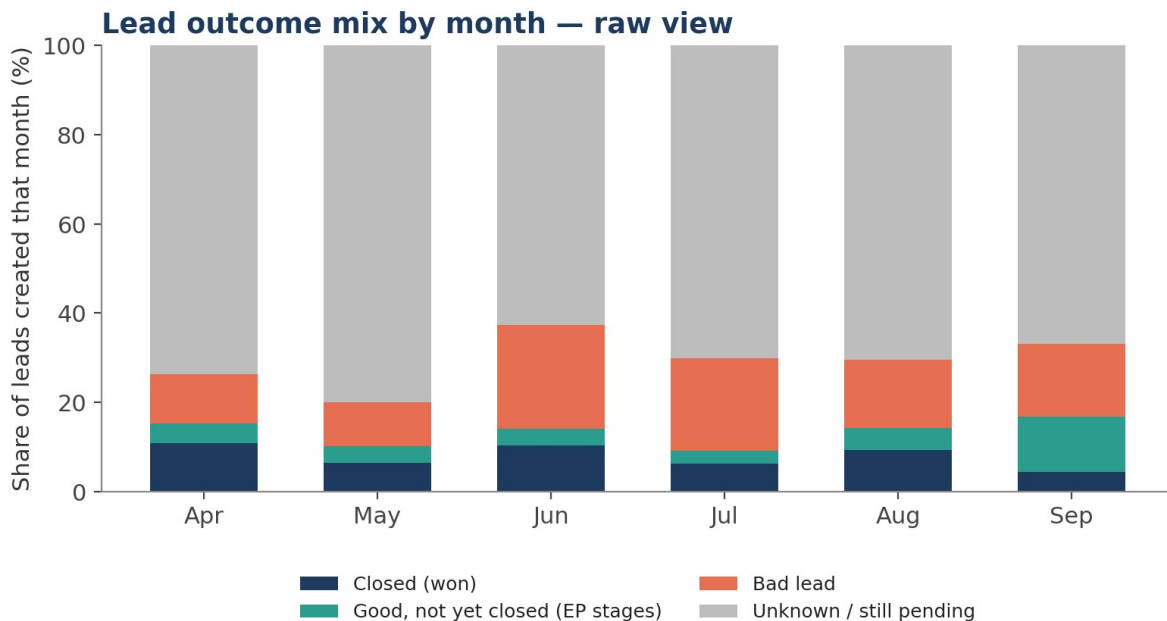


Figure 1. Outcome mix by month, all leads. Gray = still pending; this is large and uneven across months, so raw Closed-rate-by-month is not yet a clean read on quality.

1.2 Looking only at leads that have already resolved

Restricting to the 881 leads with a final status removes most of the pending-leads noise. The pattern is clear and statistically significant (chi-square across months, $p < 0.001$):

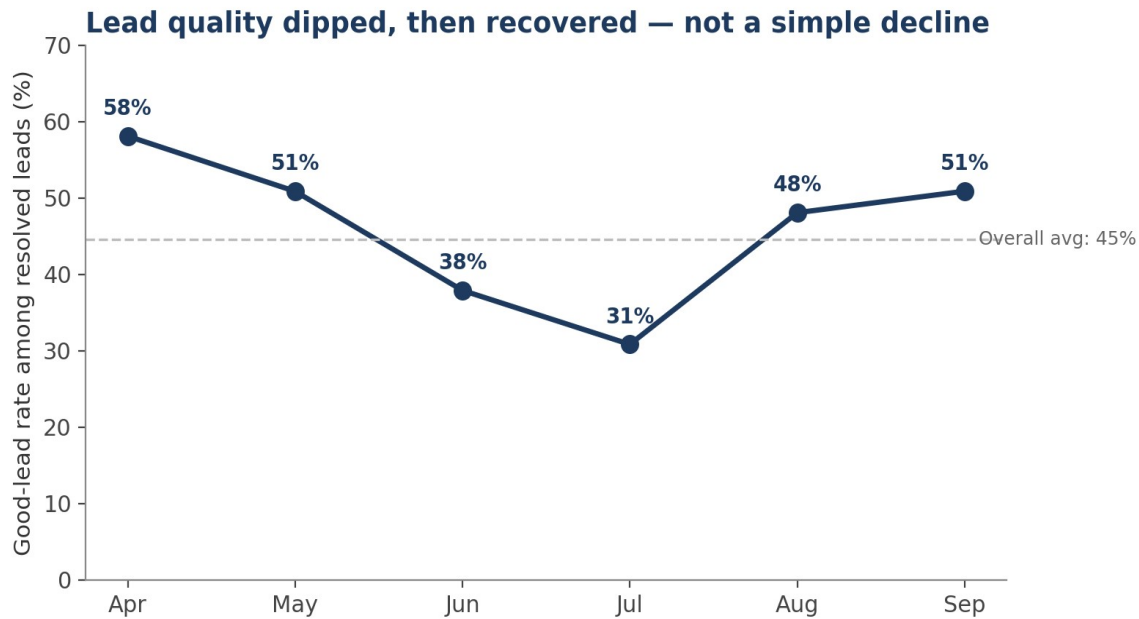


Figure 2. "Good lead" rate (Closed + EP-stage leads) among leads with a final status, by month.

Quality fell from **58% in April to 31% in July**, then recovered to **48–51% by August–September**. A logistic regression of outcome on time fits this curved (quadratic) pattern far better than a straight-line trend (likelihood-ratio test, $p < 0.0001$) — this is not noise, and it is not a one-directional decline or improvement.

1.3 What's driving the dip

Traffic-source mix shifted materially over the same window: Google's share of lead volume jumped from ~44–49% (April–June) to 70% in July, while Yahoo and AdKnowledge — both higher-quality sources (see Section 2) — lost share. The mix normalized again in August–September, tracking the quality recovery almost exactly:

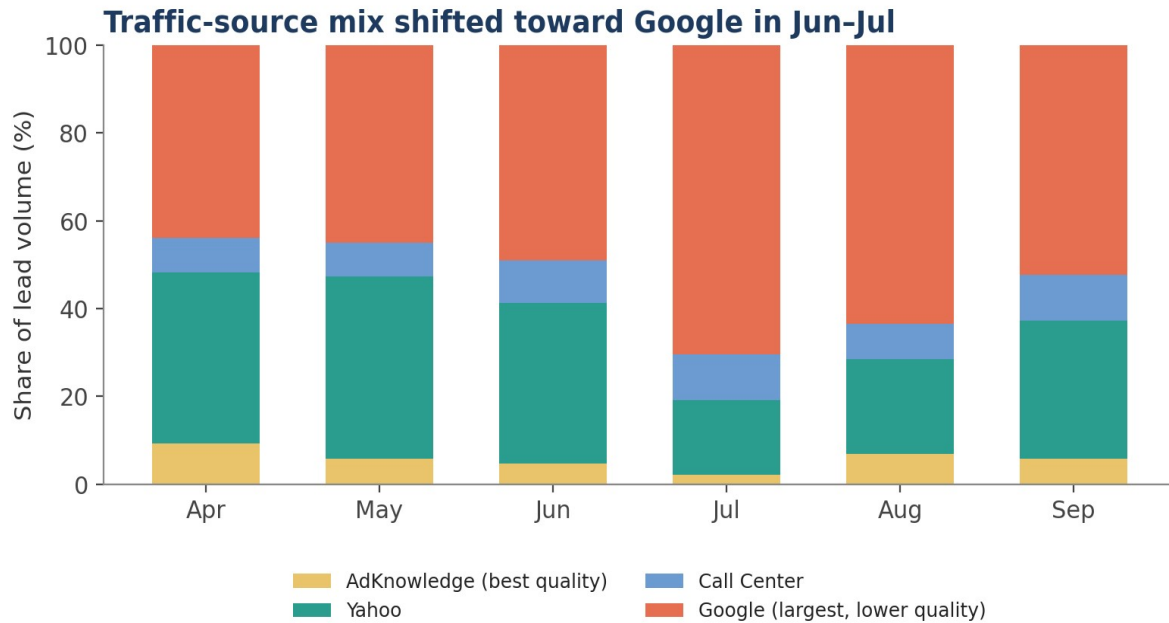


Figure 3. Share of lead volume by traffic source, by month.

A multivariate model that controls for both traffic-source mix and debt-level mix still finds a significant residual time pattern ($p < 0.0001$ on the curvature term) — so the traffic shift explains *part* of the dip, but not all of it. The remaining portion is worth investigating directly with the publisher/Google buying team (e.g., a change in placements, a fraud/bot wave, or a shift in keyword targeting during June–July) — the data here can flag the pattern but can't pinpoint the root cause without campaign-level logs.

Bottom line for Q1: the apparent “trend” is real and statistically significant, but it's a recoverable dip tied substantially to traffic-source mix — not evidence of a structurally worsening (or improving) lead pool.

2. What Drives Lead Quality?

Each candidate driver in the dataset was tested against both the Closed rate and the broader Good-lead rate. Two factors stand out as real, statistically supported drivers; several others tested show no detectable effect.

2.1 Debt level — the strongest, most actionable driver

Self-reported debt level has a clear inverted-U relationship with quality (chi-square $p = 0.006$). Leads with **too little debt (under \$10k)** structurally don't qualify for the advertiser's debt-relief program, and leads with **very high debt (over \$100k)** close at the lowest rate of any segment — plausibly seen as too large a problem, or a worse credit-risk profile, for the program. The middle bands (\$10k–\$90k) closed at 7–14%, comfortably above the 8.1% overall average.

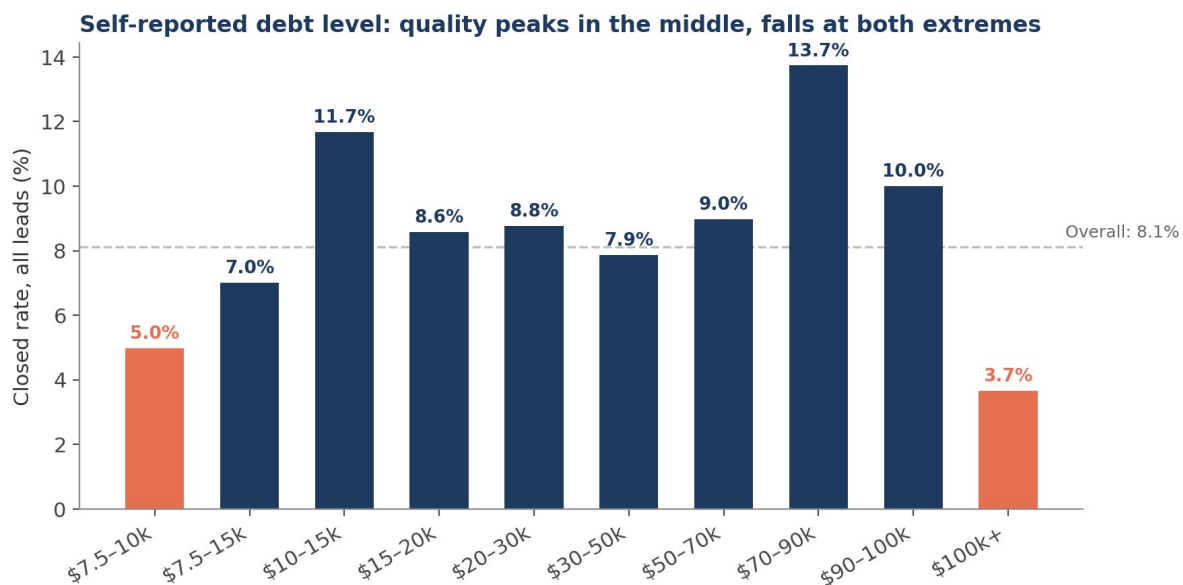


Figure 4. Closed rate by self-reported debt level, all leads.

2.2 Traffic source — a second, meaningful lever

AdKnowledge (12.3% closed, $n=171$) and the Call Center channel (9.6%, $n=271$) outperform Yahoo (7.7%) and especially Google (7.6%) — the largest-volume source. A multivariate model controlling for debt mix and time confirms Google leads close at a significantly lower rate than AdKnowledge leads ($p = 0.024$); the gap for Call Center is borderline ($p = 0.053$) and for Yahoo is directionally similar but not significant on its own. Given AdKnowledge and Call Center are also the smaller-volume sources here, this should be read as a promising, not yet fully proven, lever — worth testing with a deliberate volume reallocation.

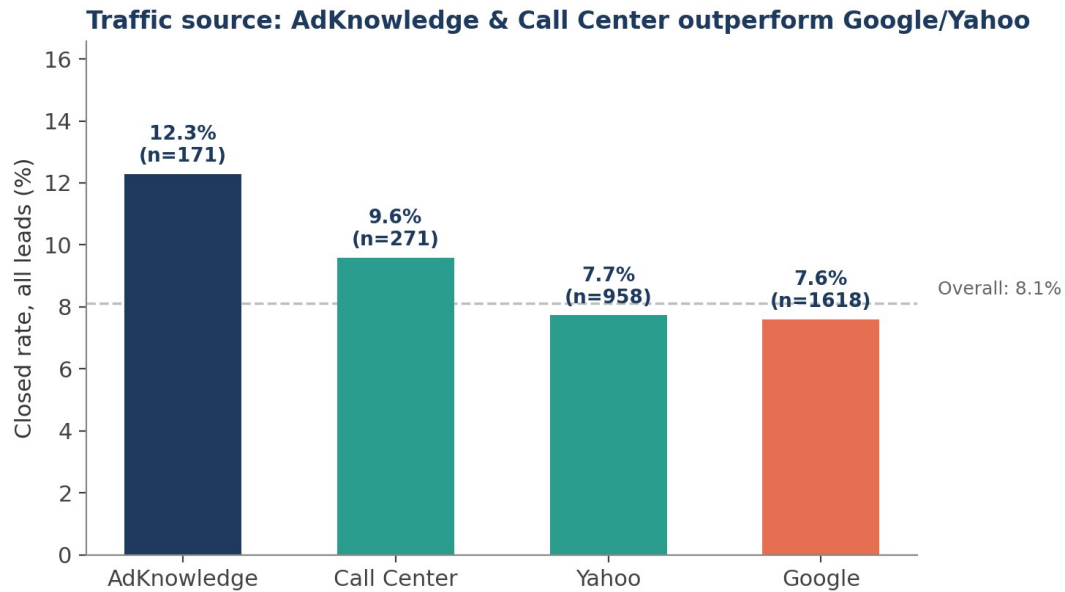


Figure 5. Closed rate by traffic source, all leads.

2.3 Address/phone verification — promising, early

Among the July–September leads where this data exists, leads where both the address and phone matched the consumer's name closed at a meaningfully higher rate. The gap is directionally consistent across both the broad “Good lead” measure ($p = 0.01$) and the stricter Closed-only measure, though the Closed-only gap isn't yet statistically conclusive given only three months of data and a smaller sample:

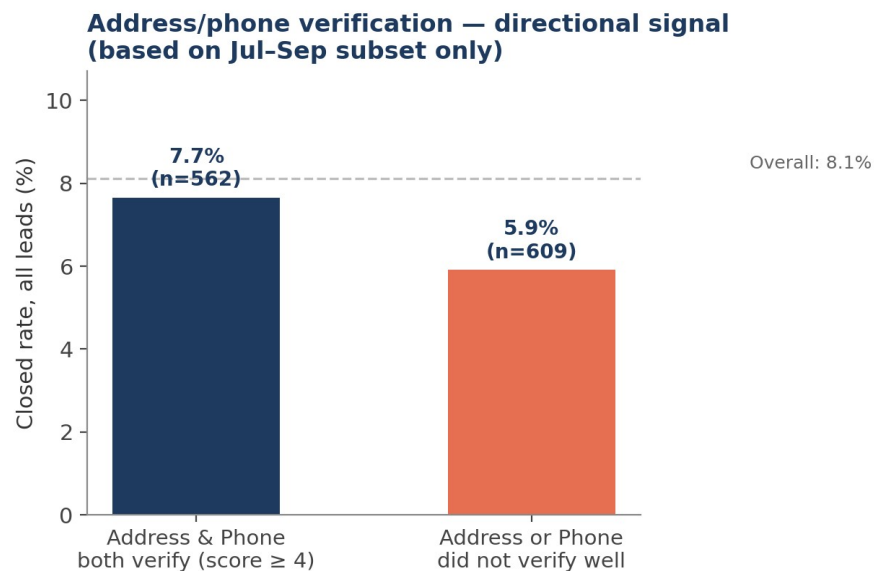


Figure 6. Closed rate by address/phone verification result, leads with score data (Jul–Sep only).

2.4 What did not move the needle

- **Ad creative / widget design:** No significant difference between the branded “CreditSolutions” design and generic designs ($p = 0.81$), and no significant difference by

background color/layout variant. Some individual widgets show large swings (e.g., one variant at 78% good-rate vs. another at 38%), but each has only 18–25 resolved leads — far too few to separate a real creative effect from noise. This is worth a properly powered A/B test, not a conclusion from this dataset.

- **Form length (1-page vs. 2-page):** No significant difference ($p = 0.70$).
- **Branded vs. generic ad copy (AdvertiserCampaignName):** No significant difference ($p = 0.81$).
- **Call-center vs. online form entry:** No significant difference ($p = 0.35$) — useful to know, since the channels likely carry very different costs to operate.
- **State:** Good-rates range widely (22%–59%) across states, but per-state samples are small (most under 50 leads); this is more likely sampling noise than a real geographic effect and wasn't formally significant.

Bottom line for Q2: debt level and traffic source are the two levers backed by the data. Verification scores are a promising third lever pending more data. Creative, form design, and channel (call center vs. web) are not shown to matter here.

3. Can We Get From 8.0% to 9.6% Lead Quality?

Yes — and the data suggests it's achievable through multiple, combinable levers rather than one big swing. The table below sizes each lever using the historical data:

Lever	Segment	Historical closed rate	Sample size
Do nothing (baseline)	All leads	8.1%	n = 3,021
Qualify on debt level	Exclude <\$10k and >\$100k debt	9.0%	n = 2,298
Shift traffic mix	AdKnowledge only	12.3%	n = 171
Shift traffic mix	Call Center only	9.6%	n = 271
Combine: debt band + best sources	\$10k–\$90k debt + AdKnowledge/Call Center	11.3%	n = 344

3.1 Recommended actions

- **Add a debt-level qualifier at the point of capture.** Leads under \$10k in debt structurally don't qualify for the program and leads over \$100k close at the lowest rate of any segment. A soft form rule (warn, re-route, or simply stop selling these to this advertiser) targets the single largest, cleanest lever in the data — and on its own gets most of the way to the 9.6% target.
- **Rebalance traffic-source mix.** Increase budget/volume allocated to AdKnowledge and the Call Center channel relative to Google, where economically feasible. Even a partial shift compounds with the debt-level filter, as shown in the combined scenario above (11.3%).
- **Diagnose the June–July Google spike.** Work with the Google buying team / publisher to understand why Google's volume share roughly doubled in July, coinciding with the quality trough — and whether that surge can be avoided or filtered next time without losing the volume.
- **Build out and operationalize address/phone verification as a pre-sale filter,** not just a logged data point. It is currently only captured on recent leads (no data before July) and is a promising but unproven lever — worth a deliberate pilot with enough volume to reach statistical confidence on the Closed-rate effect specifically, not just the broader good-lead measure.
- **Hold off on creative-driven conclusions.** Don't read into current widget-level gaps (some variants show 78% vs. 38% good-rate) — sample sizes per variant (18–25 resolved leads) are too small to trust. If creative is a lever worth pursuing, run a proper A/B test with a pre-defined minimum sample per variant.
- **Don't deprioritize call-center leads to save cost without checking quality first.** This dataset shows no quality penalty for the call-center channel relative to online — so cost, not quality, should drive that channel-mix decision.

Bottom line for Q3: the levers most supported by this data — debt-level qualification and traffic-source mix — are large enough, on their own historical performance, to plausibly clear the 9.6% bar. The advertiser's offer is worth pursuing, with debt-level qualification as the fastest, lowest-risk first step.